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1. Project Overview

- Connect the React frontend to the Spring Boot backend
- Retrieve course records from the backend API
- Display course data in an HTML table
- Implement loading, error, and empty states

Submission requirements:

2. Task A — Backend Course API

2.1 CourseController

- `@RestController`
- `@RequestMapping("/api/v1/courses")`
- `GET /api/v1/courses`

2.2 Course Data

- Add at least three courses
- Course fields:
 - ID
 - Code
 - Title
 - Credits

2.3 CORS Configuration

- Allow requests from:

`http://localhost:5173`

Course Management

Total Courses: 3

ID	Code	Title	Credits
1	EWA301	Enterprise Web App	3
2	DB201	Database Systems	3
3	SE202	Software Engineering	3

3. Task B — React API Service

3.1 Axios Installation

- Install Axios using `npm install axios`

3.2 Course API Service

- Create:

```
src/services/courseApi.js
```

3.3 GET Request

- Send a GET request to:

```
http://localhost:8080/api/v1/courses
```

The screenshot shows a REST client interface with the following details:

- Method:** GET
- URL:** http://localhost:8080/api/v1/courses
- Status:** 200 OK
- Response Time:** 45 ms
- Response Size:** 450 B
- Body:** A JSON array of course data.

id	code	title	credits
0 1	EWA301	Enterprise Web App	3
1 2	DB201	Database Systems	3
2 3	SE202	Software Engineering	3

4. Task C — Course Table

4.1 Display Course Data

Display the courses in an HTML table.

4.2 Table Columns

- ID
- Code
- Title
- Credits

4.3 Course Count

Display the total number of courses using:
`courses.length`

CompusHub				
		Dashboard	Courses	Students
		Sections	Registrations	
Course Management				Total Courses: 3
ID	Code	Title	Credits	
1	EWA301	Enterprise Web App	3	
2	DB201	Database Systems	3	
3	SE202	Software Engineering	3	

5. Task D — Loading State

5.1 Loading Message

Display:

Loading courses...

while the API request is in progress.

5.2 Loading Completion

End the loading state after the request finishes using `finally`.

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Loading courses...

6. Task D — Error State

6.1 Error Handling

Display:

Unable to load courses. Check the backend and try again.

when the API request fails.

6.2 Error Test

- Stop the backend
- Refresh the React application
- Verify that the error message is displayed

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Unable to load courses. Check the backend and try again.

7. Task D — Empty State

7.1 Empty Course List

When the API successfully returns an empty array, display:

No courses available.

7.2 Empty State Condition

The empty-state message should not appear while loading or when an error occurs.

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No courses available.